

```
1 import java.util.*;
2 class Conversion
3 {
4     int n;
5     int decToBin(int decimal_number)
6     {
7         if (decimal_number == 0)
8             return 0; //base case to terminate the recursive construct
9
10        else
11
12            return (decimal_number % 2 + 10 * decToBin(decimal_number / 2));
13    }
14
15    int decToOct(int dec)
16    {
17        if(dec <= 7)
18            return dec; //base case
19
20        else
21            return (dec % 8 +(10* decToOct(dec/8))); //recursive statement to convert
22    }
23
24    String hexadecimal(int n)
25    {
26        String hexdec="";
27        int temp;
28        while (n != 0)
29        {
30            temp = n % 16;
31            if (temp < 10)
32            {
33                hexdec=((char)(temp + 48))+hexdec;
34            }
35            else
36            {
37                hexdec=((char)(temp + 55))+hexdec;
38            }
39            n/=16;
40        }
41        return hexdec;
42    }
43
44    public static void main()
45    {
46        Scanner sc=new Scanner(System.in);
47        Conversion ob=new Conversion();
48        System.out.println("Please enter a decimal number :");
```

```
49         ob.n=sc.nextInt();
50         System.out.println("1. Conversion to binary"+"\\n2.Conversion to
51 Octal"+"\\n3.Conversion to hexadecimal");
52         System.out.println("Enter your choice :"); //menu driver to execute as per
53 user's choice
54         int ch=sc.nextInt();
55         ob.work(ch,ob.n);
56     }
57
58     void work(int ch,int n)
59     {
60         switch(ch)
61         {
62             case 1:int k=decToBin(n);
63                 System.out.println("Input: "+n+"\\tConverted Output : "+k); //case 1:
64                 to convert decimal to binary
65                 break;
66
67             case 2:int m=decToOct(n);
68                 System.out.println("Input: "+n+"\\tConverted Output : "+m);// case 2:
69                 to convert decimal to octal
70                 break;
71
72             case 3:String d=hexadecimal(n);
73                 System.out.println("Input: "+n+"\\tConverted Output : "+d);// case 3:
74                 to convert decimal to hexadecimal
75                 break;
76
77             default:System.out.println("WRONG CHOICE !!");
78         }
79     }
80 }
```